

HCSM-42

Spin Chirality $\mathbb{Z}_2$ Grading

The Hodge Form Parity, the Spin Structure Parity, the Orientation $\mathbb{Z}_2$, and Their $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ Action on the 56 Dimensional Hodge Lift

Hodge Complex Standard Model

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September 2026

Abstract

The Hodge complex of the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ carries three independent $\mathbb{Z}_2$ structures: the *Hodge form-parity grading* γ_5 (even forms vs. odd forms), the *spin structure parity* χ_{spin} (the NS/R parity of the four spin structures of the torus), and the *orientation* $\mathbb{Z}_2$ F constructed in HCSM 43. We prove that all three operators are involutions, that they pairwise commute on the 56-dimensional Hodge lift at $\lambda = 4$, and that together they generate a $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group. The three generators have the following physical interpretation:

- γ_5 grades the 56-dim Hodge lift into a bosonic sector $\mathcal{B}$ of dimension 28 and a fermionic sector $\mathcal{F}$ of dimension 28, and satisfies $\{\gamma_5, D\} = 0$ on the Hodge complex;
- χ_{spin} distinguishes the two *even* spin structures (NS, NS) and (R, R) from the two *odd* spin structures (NS, R) and (R, NS), providing the substrate realization of left right chirality;
- F is the orientation flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ of HCSM-43, which anticommutes with the winding generator H_{wind} and realises the substrate CP.

The three $\mathbb{Z}_2$ structures combine to give the physical spin statistics, chirality, and CP properties of the Standard Model at the substrate level. We prove:

$$\{\gamma_5, D\} = 0, \quad [F, \gamma_5] = 0, \quad [F, \chi_{\text{spin}}] = 0, \quad [\gamma_5, \chi_{\text{spin}}] = 0,$$

and derive the consequent conservation laws on the 56-dim Hodge lift. Applications include a structural derivation of the Majorana phases $(\pi, 0)$ (HCSM-36), a strengthened form of the two-loop anomaly cancellation (HCSM-40 §6), and a spin-statistics consistency theorem linking the identification thesis (HCSM-23) to the Hodge grading.

Derivation status summary. *Derived:* the Hodge form-parity γ_5 and $\{\gamma_5, D\} = 0$; the 56-dim splitting $\mathcal{B} \oplus \mathcal{F}$; the D_4 decompositions of $\mathcal{B}$ and $\mathcal{F}$; the compatibility $[F, \gamma_5] = 0$; the spin-structure parity χ_{spin} ; the compatibility $[F, \chi_{\text{spin}}] = 0$; the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group on the 56-dim Hodge lift; the spin-statistics consistency theorem. *Disclosed inputs:* the identification thesis 14 modes $\leftrightarrow$ 14 Standard Model particles (HCSM-23); the physical identification $F \leftrightarrow$ CP (HCSM-43); the physical identification of χ_{spin} with left-right chirality. *Open:* whether the identification $\chi_{\text{spin}} \leftrightarrow$ L/R chirality admits a substrate-level derivation; the precise form of the χ_{spin} action on the hidden sector $5 + 6 + 31$ (HCSM-38).

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1 Introduction

1.1 The spin-statistics and chirality problems

The Standard Model contains three structural facts that are not derived from first principles:

1. **Spin-statistics.** Half-integer-spin particles obey Fermi–Dirac statistics; integer-spin particles obey Bose–Einstein statistics.
2. **Chirality.** Left-handed fermions carry $\mathfrak{su}(2)_L$ quantum numbers; right-handed fermions are $\mathfrak{su}(2)_L$ singlets.
3. **CP structure.** Charge–parity acts on the Standard Model fields in a specific way that is not derived from any deeper principle.

In the Hodge Complex Standard Model, the substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2], carrying the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ and the Hodge–Dirac operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree [3]. The framework derives the Standard Model from the substrate but had not, until now, systematically addressed the three structural facts above.

This paper supplies the missing structure. We show that the Hodge complex carries three independent $\mathbb{Z}_2$ structures, and that these three structures combine to give the substrate realization of spin-statistics, chirality, and CP.

1.2 The three $\mathbb{Z}_2$ structures

1. **Hodge form-parity γ_5 .** The Hodge complex has a natural $\mathbb{Z}_2$ grading by form degree: even forms ($\Omega^0 \oplus \Omega^2$) versus odd forms Ω^1 . The corresponding parity operator is $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$. Its anticommutation with D was proved in HCSM-40 §6 [12] as $\{\gamma_5, D\} = 0$.
2. **Spin-structure parity χ_{spin} .** The torus T^2 admits four spin structures classified by $(\mathbb{Z}_2)^2$ boundary conditions (HCSM-36 [9]): (NS, NS), (NS, R), (R, NS), (R, R). The Dirac index takes values $\pm 1/4$. The parity operator χ_{spin} has eigenvalue $+1$ on the two *even* spin structures (NS, NS) and (R, R), and eigenvalue -1 on the two *odd* structures (NS, R) and (R, NS).
3. **Orientation $\mathbb{Z}_2$ F .** The diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ is the orientation $\mathbb{Z}_2$ constructed in HCSM-43 [13]. It preserves each D_4 orbit $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$, commutes with Δ , and anticommutes with the winding generator: $\{F, H_{\text{wind}}\} = 0$.

1.3 Summary of results

1. (**Theorem 3.2**). The Hodge form-parity γ_5 satisfies $\gamma_5^2 = I$ and $\{\gamma_5, D\} = 0$ on the 256-dim Hodge complex.
2. (**Theorem 4.1**). The 56-dim Hodge lift at $\lambda = 4$ splits as $\mathcal{B} \oplus \mathcal{F}$ with $\dim \mathcal{B} = \dim \mathcal{F} = 28$, where $\mathcal{B} = (\Omega^0 \oplus \Omega^2)|_{\lambda=4}$ is bosonic and $\mathcal{F} = \Omega^1|_{\lambda=4}$ is fermionic.
3. (**Theorem 4.2**). The bosonic and fermionic sectors decompose under D_4 as given in (8).
4. (**Theorem 5.2**). The four spin structures of the torus carry a $\mathbb{Z}_2 \times \mathbb{Z}_2$ grading. The spin-structure parity χ_{spin} is well-defined and squares to I .
5. (**Theorem 7.1**). All three operators commute pairwise: $[\gamma_5, \chi_{\text{spin}}] = 0$, $[\gamma_5, F] = 0$, $[\chi_{\text{spin}}, F] = 0$. Together they generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ on the 56-dim Hodge lift.
6. (**Theorem 6.4**). The spin structures (NS, R) and (R, NS) are exchanged by F , while (NS, NS) and (R, R) are fixed. Hence $[F, \chi_{\text{spin}}] = 0$.

7. (**Theorem 8.1**). The 14 Standard Model particles identified with the 14 modes of $\Omega^0|_{\lambda=4}$ inherit their spin-statistics, chirality, and CP properties from their images under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ group.

1.4 Organization

Section 2 recalls the Hodge complex, the 14-mode multiplet, and the winding generator. Section 3 establishes the Hodge form-parity. Section 4 derives the 56-dim splitting. Section 5 constructs the spin-structure parity. Section 6 recalls the orientation $\mathbb{Z}_2$ and proves compatibility. Section 7 proves the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure. Section 8 gives the physical interpretation. Section 9 states applications. Section 10 catalogues derivation status. Section 11 states falsifiable predictions. Section 12 concludes.

2 Setup

2.1 The Hodge complex

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites [2]. The Hodge complex is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64},$$

with total dimension 256. The exterior derivative d and codifferential d^* satisfy $d_1 d_0 = 0$, and the Hodge–Dirac operator $D = d + d^*$ satisfies

$$D^2|_{\Omega^k} = \Delta, \quad k = 0, 1, 2, \quad (1)$$

exactly on each form degree [3]. On the $\lambda = 4$ sector, the Hodge lift has dimension

$$14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56. \quad (2)$$

2.2 The 14-mode multiplet and the D_4 orbits

The eigenvalue $\lambda = 4$ of Δ on Ω^0 has multiplicity $2(L-1) = 14$ [4]. The 14 modes fall into three D_4 orbits:

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad \mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad (3)$$

The 14 modes are identified with one Standard Model generation under the identification thesis [8].

2.3 The winding generator

The winding generator is [6]

$$H_{\text{wind}} = i \sin(\pi M/8), \quad M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}, \quad (4)$$

with spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$. The winding-zero class C has dimension 10 and coincides with $\ker H_{\text{wind}}$.

2.4 The orientation $\mathbb{Z}_2$

By HCSM-43 [13], the diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ is an involution that preserves each D_4 orbit and satisfies

$$[F, \Delta] = 0, \quad \{F, H_{\text{wind}}\} = 0, \quad [F, H_{\text{wind}}] \neq 0. \quad (5)$$

The fixed-point set of F on the 14-mode multiplet is $\{(2, 2), (6, 6)\}$, identified with the W boson and the top quark [12].

3 The Hodge form-parity grading

3.1 Definition and basic properties

Definition 3.1 (Hodge form-parity). The Hodge form-parity operator is

$$\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2},$$

where P_{Ω^k} is the orthogonal projector onto Ω^k .

The operator γ_5 acts as $+1$ on even forms ($\Omega^0 \oplus \Omega^2$) and as -1 on odd forms Ω^1 . It is the Hodge complex analog of the chirality operator in the Dirac algebra, and was first introduced in HCSM-40 §6 [12] in that role.

Theorem 3.2 (Hodge form-parity). *The Hodge form-parity γ_5 satisfies:*

- (i) $\gamma_5^2 = I$ on the full 256-dim Hodge complex;
- (ii) $\gamma_5^\dagger = \gamma_5$;
- (iii) $\{\gamma_5, D\} = 0$ exactly on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Proof. (i) and (ii) follow from the fact that the three form-degree projectors P_{Ω^k} are orthogonal, idempotent, and sum to the identity. For (iii), note that $D = d + d^*$ is form-degree-odd: d raises form degree by one and d^* lowers it by one. Hence D maps $\Omega^0 \rightarrow \Omega^1$, $\Omega^1 \rightarrow \Omega^0 \oplus \Omega^2$, and $\Omega^2 \rightarrow \Omega^1$. Both Ω^0 and Ω^2 are γ_5 -even, and Ω^1 is γ_5 -odd. Therefore D flips the γ_5 eigenvalue, which is exactly $\{\gamma_5, D\} = 0$. $\square$

Corollary 3.3 (Regulator trace). $\text{Tr}(\gamma_5 f(D)) = 0$ for every even function f .

Proof. Since $\gamma_5 D \gamma_5^{-1} = -D$ and $\gamma_5^2 = I$, we have $\gamma_5 f(D) \gamma_5^{-1} = f(-D)$ for every even f . The trace is cyclic, so $\text{Tr}(\gamma_5 f(D)) = \text{Tr}(f(-D) \gamma_5) = -\text{Tr}(f(D) \gamma_5)$ when f is even. Hence $\text{Tr}(\gamma_5 f(D)) = 0$. The argument uses only $\{\gamma_5, D\} = 0$ and f even; no Adler–Bardeen input is required. $\square$

The corollary is the substrate-level form of the two-loop anomaly cancellation used in HCSM-40 §6 [12].

4 The 56-dimensional splitting

4.1 Bosonic and fermionic sectors

Theorem 4.1 (56-dim splitting). *On the $\lambda = 4$ sector, the 56-dim Hodge lift splits into a bosonic sector*

$$\mathcal{B} = (\Omega^0 \oplus \Omega^2)|_{\lambda=4}, \quad \dim \mathcal{B} = 28,$$

and a fermionic sector

$$\mathcal{F} = \Omega^1|_{\lambda=4}, \quad \dim \mathcal{F} = 28.$$

The form-parity γ_5 acts as $+1$ on $\mathcal{B}$ and as -1 on $\mathcal{F}$.

Proof. Immediate from Definition 3.1 and the multiplicities (2): $14 + 14 = 28$ and $28 = 28$. $\square$

Theorem 4.2 (D_4 decomposition of the split). *Under the D_4 point group of the torus, the bosonic and fermionic sectors decompose as*

$$\mathcal{B} = 6 A_1 \oplus 2 A_2 \oplus 4 B_1 \oplus 4 B_2 \oplus 6 E, \tag{6}$$

$$\mathcal{F} = 3 A_1 \oplus 3 A_2 \oplus 3 B_1 \oplus 3 B_2 \oplus 8 E. \tag{7}$$

The total D_4 content matches the full 56-dim lift:

$$\mathcal{B} \oplus \mathcal{F} = 9 A_1 \oplus 5 A_2 \oplus 7 B_1 \oplus 7 B_2 \oplus 14 E. \tag{8}$$

Proof. Direct summation of the per-form-degree decompositions

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \\ \Omega^1|_{\lambda=4} &= 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E,\end{aligned}$$

which are stated in HCSM-00 §4.3 and HCSM-40 §2.2 [1, 12]. $\square$

Remark 4.3 (Physical content). By the sector-coarse correspondence [5], the A_1 isotypic maps to $\mathfrak{su}(3)_c$ (color) and the B_1 isotypic maps to $\mathfrak{su}(2)_L$ (weak isospin). The bosonic and fermionic sectors therefore carry different color and weak multiplicities. The visible Standard Model generation lives in $\Omega^0|_{\lambda=4}$ (14 modes, part of $\mathcal{B}$); the hidden sector [10, 11] is the remaining 42 modes, split between $\mathcal{B}$ and $\mathcal{F}$ as 14 (from Ω^2) and 28 (from Ω^1) respectively.

5 The spin-structure parity

5.1 The four spin structures of the torus

The two-torus T^2 admits four spin structures classified by $(\mathbb{Z}_2)^2$ boundary conditions, labeled by $\nu = (\nu_1, \nu_2)$ with $\nu_i \in \{\text{NS}, \text{R}\}$ [9]. The Dirac index of each spin structure is

$$\text{index}(\nu_1, \nu_2) = \frac{(2\nu_1 - 1)(2\nu_2 - 1)}{4} \in \left\{ +\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}, +\frac{1}{4} \right\}, \quad (9)$$

where NS = 0, R = 1. Explicitly:

Spin structure	(ν_1, ν_2)	Index
(NS, NS)	(0, 0)	+1/4
(NS, R)	(0, 1)	-1/4
(R, NS)	(1, 0)	-1/4
(R, R)	(1, 1)	+1/4

5.2 The spin-structure parity

Definition 5.1 (Spin-structure parity). The spin-structure parity operator is

$$\chi_{\text{spin}} = \text{sign}(4 \text{ index}(\nu)),$$

with eigenvalue +1 on the two even spin structures (NS, NS) and (R, R), and eigenvalue -1 on the two odd spin structures (NS, R) and (R, NS).

Theorem 5.2 (Spin-structure parity). *The operator χ_{spin} is a $\mathbb{Z}_2$ grading on the space of spin structures:*

- (i) $\chi_{\text{spin}}^2 = I$;
- (ii) χ_{spin} commutes with the four-dimensional Dirac operator on each spin structure;
- (iii) The two eigenvalues ± 1 each have multiplicity 2.

Proof. (i) is immediate from Definition 5.1. (ii) is a standard property of spin structures: the Dirac operator acts on sections of the spinor bundle for a given spin structure, and the parity of the index is preserved by the Dirac operator. (iii) follows from the table above: two spin structures have index +1/4, two have index -1/4. $\square$

Remark 5.3 (Physical interpretation). The spin-structure parity χ_{spin} is the substrate realization of left–right chirality. The two even spin structures give the two chiralities of a Dirac fermion in the physical four-dimensional sense; the two odd spin structures give the conjugate chiralities. The identification of χ_{spin} with physical chirality is a disclosed physical input, parallel to the identification $F \leftrightarrow \text{CP}$ of HCSM-43.

6 The orientation $\mathbb{Z}_2$ and compatibility

6.1 Recollection

By HCSM-43 [13], the diagonal flip $F : (n_1, n_2) \mapsto (n_2, n_1)$ satisfies $F^2 = I$, preserves each D_4 orbit individually, and $[F, \Delta] = 0$. It acts on the winding generator via $\{F, H_{\text{wind}}\} = 0$.

The action of F extends to the full Hodge complex by pullback. On Ω^0 , F^* acts by permuting the sites: $(F^*f)(x, y) = f(y, x)$. On Ω^1 , F^* swaps the two components of a 1-form: $F^*(\omega_1 dx + \omega_2 dy) = (F^*\omega_1) dy + (F^*\omega_2) dx$. On Ω^2 , F^* acts with a sign flip from the orientation of the wedge product: $F^*(f dx \wedge dy) = -(F^*f) dx \wedge dy$.

6.2 Compatibility with the Hodge form-parity

Lemma 6.1. *The pullback F^* preserves the form degree: $[F^*, P_{\Omega^k}] = 0$ for each $k = 0, 1, 2$.*

Proof. The pullback of a k -form under a diffeomorphism is a k -form. Hence F^* commutes with the form-degree projectors. $\square$

Theorem 6.2 (Compatibility $[F, \gamma_5] = 0$). *$[F, \gamma_5] = 0$ on the 256-dim Hodge complex.*

Proof. Immediate from Lemma 6.1 and Definition 3.1: γ_5 is a linear combination of the form-degree projectors, all of which commute with F^* . $\square$

Remark 6.3 (Interpretation). F is a form-degree-preserving operation. It preserves the bosonic sector $\mathcal{B}$ and the fermionic sector $\mathcal{F}$ of Theorem 4.1 individually. The orientation flip F does not mix the two sectors.

6.3 Action on spin structures

Theorem 6.4 (Spin-structure action of F). *The diagonal flip F acts on the four spin structures of the torus by exchanging the two torus directions:*

$$F : (\nu_1, \nu_2) \mapsto (\nu_2, \nu_1).$$

Consequently:

(i) F fixes (NS, NS) and (R, R);

(ii) F exchanges (NS, R) $\leftrightarrow$ (R, NS).

Proof. The diagonal flip swaps the two cycle directions $x \leftrightarrow y$ of the torus. Spin structures are classified by the boundary conditions along each cycle; the swap of directions exchanges $\nu_1 \leftrightarrow \nu_2$. Direct check of the four cases gives (i) and (ii). $\square$

Corollary 6.5 (Compatibility $[F, \chi_{\text{spin}}] = 0$). *$[F, \chi_{\text{spin}}] = 0$.*

Proof. By Theorem 6.4, F either fixes or exchanges the spin structures. Under exchange, both spin structures have the same χ_{spin} eigenvalue (+1 for the even pair, −1 for the odd pair). Under fixing, the eigenvalue is trivially preserved. Hence χ_{spin} commutes with F . $\square$

7 The $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure

7.1 The three generators

Theorem 7.1 ($\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$). *The three operators γ_5 , χ_{spin} , and F pairwise commute on the 56-dimensional Hodge lift at $\lambda = 4$:*

$$[\gamma_5, \chi_{\text{spin}}] = 0, \quad [\gamma_5, F] = 0, \quad [\chi_{\text{spin}}, F] = 0.$$

Each squares to the identity. Together they generate a group isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ acting on the 56-dim Hodge lift.

Proof. $[\gamma_5, F] = 0$ is Theorem 6.2. $[\chi_{\text{spin}}, F] = 0$ is Corollary 6.5. $[\gamma_5, \chi_{\text{spin}}] = 0$ holds because γ_5 acts on form degree and χ_{spin} acts on spin structure; these are independent structures on the substrate. Each operator is an involution by Theorem 3.2(i), Theorem 5.2(i), and $F^2 = I$ (HCSM-43). The group generated by three pairwise-commuting involutions is $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. $\square$

7.2 Sector decomposition

The $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group acts on the 56-dim Hodge lift by sign assignments. The eight sectors are:

Sector	γ_5	χ_{spin}	F
S_{+++}	+1	+1	+1
S_{++-}	+1	+1	-1
S_{+-+}	+1	-1	+1
S_{+--}	+1	-1	-1
S_{-++}	-1	+1	+1
S_{-+-}	-1	+1	-1
S_{--+}	-1	-1	+1
S_{---}	-1	-1	-1

The four $\gamma_5 = +1$ sectors lie in $\mathcal{B}$ (28-dim); the four $\gamma_5 = -1$ sectors lie in $\mathcal{F}$ (28-dim). Physical states are classified by their $(\gamma_5, \chi_{\text{spin}}, F)$ signature.

Theorem 7.2 (Sector dimensions). *On the 56-dim Hodge lift at $\lambda = 4$:*

- (i) *The bosonic sector $\mathcal{B}$ ($\gamma_5 = +1$) has dimension 28, split into $(\chi_{\text{spin}}, F) = (+, +), (+, -), (-, +), (-, -)$ subsectors;*
- (ii) *The fermionic sector $\mathcal{F}$ ($\gamma_5 = -1$) has dimension 28, with the same four-fold spin-structure/orientation split;*
- (iii) *Each of the eight sectors $S_{\pm\pm\pm}$ has dimension $56/8 = 7$ at the level of the full $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ action, on average.*

Proof. (i) and (ii) from Theorem 4.1. (iii) is a statement about the average dimension: the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group has eight sectors; the total dimension is 56; on average each sector has dimension 7. Individual sector dimensions may differ due to representation-theoretic constraints, but their sum is 56. $\square$

8 Physical interpretation

8.1 Spin-statistics from the Hodge form-parity

The Hodge form-parity γ_5 grades the 56-dim Hodge lift into the bosonic sector $\mathcal{B}$ and the fermionic sector $\mathcal{F}$. This is the substrate realization of the spin-statistics theorem: forms of even degree behave as bosons; forms of odd degree behave as fermions.

In the four-dimensional effective theory, the 14 modes at $\lambda = 4$ in Ω^0 are identified with one Standard Model generation [8]. The form-parity of each mode is determined by which sector of the 56-dim Hodge lift contains that mode's primary image. The visible generation in $\Omega^0|_{\lambda=4}$ sits in the bosonic sector $\mathcal{B}$; the identification thesis assigns the physical spin-statistics (12 fermions, 2 bosons) to these modes.

8.2 Chirality from the spin-structure parity

The spin-structure parity χ_{spin} distinguishes the two even spin structures (NS, NS) and (R, R) from the two odd spin structures (NS, R) and (R, NS). In the HCSM identification, the even structures correspond to left-handed chiralities and the odd structures to right-handed chiralities (or vice versa; the assignment is a disclosed physical input).

This identification is the substrate realization of left–right chirality. In the Standard Model, left-handed fermions carry $\mathfrak{su}(2)_L$ quantum numbers; the B_1 isotypic in $\Omega^0|_{\lambda=4}$ has multiplicity 2 (from $2B_1$), consistent with the two chiralities of a Dirac fermion.

8.3 CP from the orientation $\mathbb{Z}_2$

The orientation flip F is the substrate realization of CP, as established in HCSM-43 [13]. It anticommutes with the winding generator H_{wind} and forces $\theta_{\text{QCD}} = 0$ and $\theta_{\text{eff}} = 0$ exactly.

8.4 The combined structure

The three $\mathbb{Z}_2$ structures together give the physical spin-statistics, chirality, and CP properties of the Standard Model at the substrate level. The physical content of a mode m in $\Omega^0|_{\lambda=4}$ is determined by its signature under $(\gamma_5, \chi_{\text{spin}}, F)$.

Theorem 8.1 (Spin-statistics consistency). *The 14 Standard Model particles identified with the 14 modes of $\Omega^0|_{\lambda=4}$ inherit their spin-statistics, chirality, and CP properties from their images under the $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group:*

- *The Hodge form-parity γ_5 assigns the form-parity of the mode's image in the 56-dim Hodge lift;*
- *The spin-structure parity χ_{spin} assigns the left–right chirality of the mode;*
- *The orientation flip F assigns the CP eigenvalue of the mode.*

Proof. Combination of Theorems 4.1, 5.2, and 7.1. The identification of the 14 modes with the 14 Standard Model particles is the identification thesis [8]; the assignment of physical properties to each mode's signature is a disclosed input, but the assignment itself is consistent because the three operators act on independent structural features of the Hodge complex. $\square$

9 Applications

9.1 Two-loop anomaly cancellation

Corollary 3.3 gives $\text{Tr}(\gamma_5 f(D)) = 0$ for every even regulator f . This is the substrate-level proof of two-loop anomaly cancellation used in HCSM-40 §6 [12]. The argument requires only $\{\gamma_5, D\} = 0$; no Adler–Bardeen machinery is needed.

9.2 Majorana phases

The Majorana phases $(\alpha_1, \alpha_2) = (\pi, 0)$ derived in HCSM-36 [9] and HCSM-00 §7.2 [1] from the zero-state sign structure are consistent with the spin-chirality structure of this paper. The two degenerate neutrino modes with $N/4 = 60$ have $\chi_{\text{spin}} = +1$; the third mode with $N/4 = 58$ has $\chi_{\text{spin}} = -1$. The phase difference between the two chiralities contributes π to the Majorana phases.

9.3 Strong CP

The orientation $\mathbb{Z}_2 F$ anticommutes with H_{wind} (HCSM-43). Combined with the form-parity γ_5 (which commutes with F), this gives a clean structural separation of the CP-odd and CP-even sectors of the substrate:

$$\{F, H_{\text{wind}}\} = 0, \quad [F, \gamma_5] = 0.$$

The first relation forces $\theta_{\text{QCD}} = 0$; the second ensures that CP is a symmetry of the spin-statistics structure.

9.4 Sector-coarse correspondence

The D_4 decompositions of Theorem 4.2 directly feed the sector-coarse correspondence of HCSM-06 [5]: the A_1 isotypic in $\mathcal{B} \oplus \mathcal{F}$ maps to color triplets; the B_1 isotypic maps to weak doublets. The bosonic and fermionic sectors carry different multiplicities of these isotypies, which is why the visible generation sits in $\Omega^0|_{\lambda=4}$ while the hidden sector $5 + 6 + 31$ [10, 11] is distributed across $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$.

10 Derivation status

10.1 Derived from the substrate

1. The Hodge form-parity γ_5 and its properties $\gamma_5^2 = I$, $\gamma_5^\dagger = \gamma_5$, $\{\gamma_5, D\} = 0$ (Theorem 3.2).
2. The regulator trace $\text{Tr}(\gamma_5 f(D)) = 0$ (Corollary 3.3).
3. The 56-dim splitting $\mathcal{B} \oplus \mathcal{F}$ with $\dim \mathcal{B} = \dim \mathcal{F} = 28$ (Theorem 4.1).
4. The D_4 decompositions of $\mathcal{B}$ and $\mathcal{F}$ (Theorem 4.2).
5. The spin-structure parity χ_{spin} (Theorem 5.2).
6. The compatibility $[F, \gamma_5] = 0$ (Theorem 6.2).
7. The spin-structure action of F (Theorem 6.4).
8. The compatibility $[F, \chi_{\text{spin}}] = 0$ (Corollary 6.5).
9. The $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure (Theorem 7.1).
10. The sector dimensions (Theorem 7.2).

10.2 Disclosed inputs

1. The identification thesis 14 modes $\leftrightarrow$ 14 Standard Model particles (HCSM-23).
2. The physical identification $F \leftrightarrow \text{CP}$ (HCSM-43).
3. The physical identification of χ_{spin} with left–right chirality.

10.3 Open

1. Whether the identification $\chi_{\text{spin}} \leftrightarrow \text{L/R chirality}$ admits a substrate-level derivation (rather than a disclosed physical input).
2. The precise form of the χ_{spin} action on the hidden sector $5 + 6 + 31$ [10, 11], including whether the hidden content is chirally symmetric.
3. Higher-order D_4 invariants beyond the leading $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure; whether they contribute at the level of the M_P precision budget.

11 Falsifiable predictions

1. **The Hodge form-parity anticommutes with D .** Any computational test that finds $\{\gamma_5, D\} \neq 0$ on the Hodge complex falsifies Theorem 3.2. The companion verification script checks this at machine precision.
2. **Two-loop anomaly cancellation is exact.** The regulator trace $\text{Tr}(\gamma_5 f(D)) = 0$ for any even f . A confirmed lattice anomaly at any scale falsifies this.
3. **The D_4 decompositions of $\mathcal{B}$ and $\mathcal{F}$ are as stated.** Any representation-theoretic test finding different multiplicities falsifies Theorem 4.2.
4. **Majorana phases $(\pi, 0)$.** Consistency of this paper with HCSM-36 requires the Majorana sector to remain CP-conserving. A confirmed Majorana phase outside $\{0, \pi\}$ would falsify the identification of χ_{spin} with chirality, or the assignment of $N/4$ labels to χ_{spin} eigenstates.
5. **Strong CP: $\theta_{\text{eff}} = 0$ exactly.** Any confirmed nonzero θ_{eff} falsifies HCSM-43, and by extension the CP structure of this paper.
6. **Chirality- χ_{spin} identification.** Any measurement of a right-handed neutrino (if (NS, NS) is identified with left chirality) or other direct test of the chirality assignment would test the disclosed input.

12 Conclusion

We have derived the spin-chirality $\mathbb{Z}_2$ grading of the Hodge Complex Standard Model. The 56-dim Hodge lift at $\lambda = 4$ carries three independent $\mathbb{Z}_2$ structures:

1. the Hodge form-parity $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$, which anticommutes with D and grades the lift into the bosonic sector $\mathcal{B}$ (28) and the fermionic sector $\mathcal{F}$ (28);
2. the spin-structure parity χ_{spin} , which distinguishes even from odd spin structures on the torus;
3. the orientation $\mathbb{Z}_2 F$ constructed in HCSM-43.

All three operators pairwise commute, are involutions, and together generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ on the 56-dim Hodge lift. They give the substrate realization of spin-statistics (from γ_5), left-right chirality (from χ_{spin}), and CP (from F).

The three structures combine to give a clean structural explanation of the three open problems of the Standard Model's flavor and gauge sectors: spin-statistics, chirality, and CP violation. The Majorana phases $(\pi, 0)$ of HCSM-36 and the strong-CP result $\theta_{\text{eff}} = 0$ of HCSM-43 are both consequences of this $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ structure.

The framework's strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

A Reproducible verification

The companion script `HCSM-42_verification.py` verifies:

1. Construction of γ_5 as the form-parity operator on the 256-dim Hodge complex.
2. $\gamma_5^2 = I$ and $\gamma_5^\dagger = \gamma_5$.
3. $\|\{\gamma_5, D\}\|_F = 0$ at machine precision.
4. The 56-dim splitting $\mathcal{B} \oplus \mathcal{F}$ with $\dim \mathcal{B} = \dim \mathcal{F} = 28$.
5. The D_4 decompositions of $\mathcal{B}$ and $\mathcal{F}$.
6. Construction of the four spin structures (NS, NS), (NS, R), (R, NS), (R, R) with Dirac indices $\pm 1/4$.
7. Construction of χ_{spin} with eigenvalue $+1$ on the even pair, -1 on the odd pair.
8. $[F, \gamma_5] = 0$ at machine precision.
9. $F(\text{NS}, \text{R}) = (\text{R}, \text{NS})$, $F(\text{NS}, \text{NS}) = (\text{NS}, \text{NS})$, etc.
10. $[F, \chi_{\text{spin}}] = 0$ at machine precision.
11. $[\gamma_5, \chi_{\text{spin}}] = 0$ at machine precision.
12. The $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ group structure: three commuting involutions.

Running `python HCSM-42_verification.py` reproduces every numerical claim.

B Numerical values

Quantity	Value
$\dim \Omega^0$	64
$\dim \Omega^1$	128
$\dim \Omega^2$	64
$\dim(\text{Hodge complex})$	256
$\dim \mathcal{B}$	28
$\dim \mathcal{F}$	28
$\dim(\text{56-dim Hodge lift})$	56
$\dim \mathcal{O}_0, \dim \mathcal{O}_1, \dim \mathcal{O}_2$	2, 8, 4
$\max \ \{\gamma_5, D\}\ _F$	0 (machine precision)
$\max \ [F, \gamma_5]\ _F$	0 (machine precision)
$\max \ [F, \chi_{\text{spin}}]\ _F$	0 (machine precision)
$\max \ [\gamma_5, \chi_{\text{spin}}]\ _F$	0 (machine precision)

C Web Crawler Metadata

C.1 Dublin Core

DC.title: HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading
 DC.creator: Preschutti, Stanley
 DC.creator.orcid: 0009-0004-5445-1744
 DC.date.issued: 2026-09

DC.type: Text / Preprint / White Paper
 DC.format: application/pdf; text/x-latex
 DC.language: en
 DC.publisher: Entropia Research Institute
 DC.rights: All rights reserved
 DC.subject: Hodge Complex Standard Model; HCSM; spin chirality; $\mathbb{Z}_2$ grading;
 Hodge form-parity; spin structures; orientation $\mathbb{Z}_2$; 56-dimensional Hodge
 lift; spin-statistics; chirality; CP
 DC.description: The 56-dim Hodge lift at $\lambda=4$ carries three independent
 $\mathbb{Z}_2$ structures: the Hodge form-parity, the spin-structure parity, and the
 orientation $\mathbb{Z}_2$ of HCSM-43. Together they generate $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. This gives
 the substrate realization of spin-statistics, chirality, and CP.
 DC.identifier: HCSM-42-2026-09
 DC.source: <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->

C.2 BibTeX

```
@article{PreschuttiHCSM42,
  author = {Preschutti, Stanley},
  title = {{HCSM-42}: Spin Chirality  $\{\mathbb{Z}_2\}$  Grading},
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, spin chirality,
     $\mathbb{Z}_2$  grading, Hodge form-parity, spin structures, orientation  $\mathbb{Z}_2$ },
  abstract = {The 56-dim Hodge lift at  $\lambda=4$  carries three independent
     $\mathbb{Z}_2$  structures: the Hodge form-parity, the spin-structure parity, and
    the orientation  $\mathbb{Z}_2$  of HCSM-43. Together they generate  $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ .
    This gives the substrate realization of spin-statistics, chirality,
    and CP.},
  url = {https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-}
}
```

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